General Physics(Phys1001)

(Chap- 1)

Department of Physics

Addis Ababa Science and Technology University

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Course Outline

1. **Preliminaries (Physical Quantities)**
2. **Kinematics and Dynamics of Particle**
3. **Fluids Mechanics**

Test : chap1-2(25%)

4. **Heat and Thermodynamics**
5. **Oscillations, Waves and Optics**
6. **Electromagnetism and Electronics**
7. **Cross Cutting Applications of Physics**

Final Exam: chap1-7 (50%)

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Evaluation

- Demonstration/Lab work **(10%)**
- Quizzes (5%)
- Assignments **(10 %)**
- **Test (25 %)**
- Final examination(**50%**

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Course Information: Materials

- Textbook:

  Physics for Scientists and Engineers", 6th----10th Edition, by Serway and Jewett

- Reference books:

  - *are given in the course outline*

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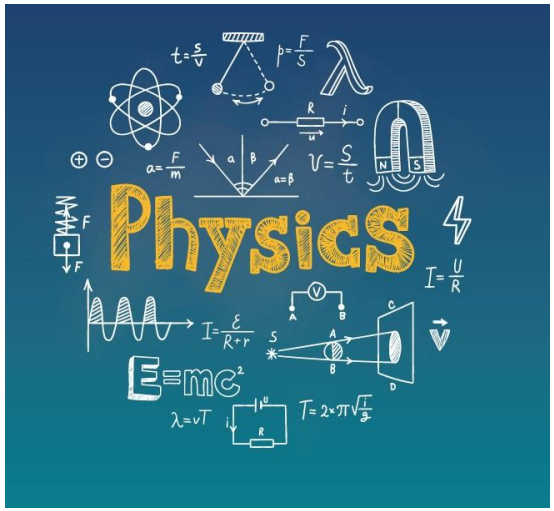
CHAPTER ONE

PRELIMINARIES

- The word physics comes from the Greek word meaning “nature”. Today physics is treated as the base for science and have various applications for the ease of life.



- Physics deals with matter in relation to energy and the accurate measurement of natural phenomenon.

- Thus physics is a science of measurement.

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Learning Objectives: At the end of this chapter, you will be able to:

- Explain physics.
- Describe how SI base units are defined.
- Describe how derived units are created from base units.
- Express quantities given in SI units using metric prefixes.
- Describe the relationships among models, theories, and laws.
- Know the units used to describe various physical quantities.
- Become familiar with the prefixes used for larger and smaller quantities.
- Master the use of unit conversion (dimensional analysis) in solving problems.
- Understand the relationship between uncertainty and the number of significant figures in a number.

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physical quantity

Length of a car is 4.5 m

physical quantity magnitude

Physical quantity is a quantifiable or assignable property ascribed to a particular phenomenon or body, for instance the length of a rod or the mass of a body.

Measurement is the act of comparing a physical quantity with a certain standard.

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A. Basic Physical Quantities:

Basic quantities are the quantities which cannot be expressed in terms of any other physical quantity. Example: length, mass and time.

B. Derived Physical Quantities:

Derived quantities are quantities that can be expressed in terms of fundamental quantities. Examples: area, volume, density.

1.1.2. SI Units: Basic and Derived Units

SI unit is the abbreviation for **International System of Units** and is the modern form of metric systemfinallyagreeduponattheeleventhInternationalconferenceofweightsandmeasures,1960. This

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Table 1.1: Basic quantities and their SI units

Property	Symbol	Unit	Dimension
Length	L	meter (m)	L
Mass	m	kilogram (kg)	M
Time	t	second (s)	T
Temperature	T	kelvin (K)	θ
Electric Current	I	ampere (A)	I
Amount of Substance	N	mole (N)	1
Luminous Intensity	F	candela (cd)	J

Table 1.2: Derived quantities, their SI units and dimensions

Property	Symbol	Unit	Dimension
Force	F	newton (N)	$\mathrm{kg{\cdot}m{\cdot}s^{-2}} = \mathrm{kg{\cdot}m{\cdot}/s^2}$
Speed	v	meter per second (m/s)	$\mathrm{m{\cdot}s^{-1}} = \mathrm{m/s}$
Pressure	P	pascal (Pa)	(force per unit area) $\mathrm{kg{\cdot}m^{-1}{\cdot}s^{-2}}$
Energy	E	joule (J)	$\mathrm{kg{\cdot}m^2{\cdot}s^{-2}}$
Power	W	watt (W)	(energy per unit time) $\mathrm{kg{\cdot}m^2{\cdot}s^{-3}}$

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1.1.3. Conversion of Units

Quantity	From	To	Operation
Length	inch (in)	m	(inch) $\times$ 0.0254
	foot (ft)	m	(foot) $\times$ 0.3048
	mile (mi)	m	(mile) $\times$ 1609.34
Mass	pound (lb)	kg	(pound) $\times$ 0.4536
	metric ton (t)	kg	(ton) $\times$ 1000
	ounce	kg	(ounce) $\times$ 0.02835
Volume	liter (l)	m^3	$(liter) \times 0.001$
	gallon (ga)	m^3	(gallon) $\times$ 0.00379
Temperature	fahrenheit (F)	K	$\{(\text{fahrenheit}) - 32\} \times \frac{5}{9} + 273.15$
	celcius (C)	K	(celcius) + 273.15

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Activities:

1. A common Ethiopian cities speed limit is 30km/hr. What is this speed in miles per hours?
2. How many cubic meters are in 250,000 cubic centimeters?
3. The average body temperature of a house cat is 101.5oF. What is this temperature in Celsius?

1.2. Uncertainty in Measurement and Significant Digits

1. **Systematic Error** - errors resulting from measuring devices being out of calibration. Such measurements will be consistently too small or too large. These errors can be eliminated by pre-calibrating against a known, trusted standard.
2. **Random Errors** - errors resulting in the fluctuation of measurements of the same quantity about the average. The measurements are equally probable of being too large or too small. These errors generally result from the fineness of scale division of a measuring device.

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1. Uncertainty in a scale measuring device is equal to the smallest increment divided by 2.

$$\sigma_x = \frac{Smallest\ increament}{2}$$

Example: Meter Stick (scale device)

$$\sigma_x = \frac{1mm}{2} = 0.5mm = 0.05cm$$

2. Uncertainty in a digital measuring device is equal to the smallest increment.

$$\sigma_x = smallest\ increament$$

Example: A reading from digital Balance (digital device) is 5.7513 kg, therefore

$$\sigma_x = 0.0001$$

$Measurement = x_{best} \pm \sigma_x$

Where, x_{best}= best estimate of measurement, σ_x= uncertainty (error) in measurement.

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1.2.1. Significant digits

In general to determine significant digits in a given number

1. All non-zero numbers are significant.
2. Zeros within a number are always significant.
3. Zeros that do nothing but set the decimal point are not significant. Both 0.000098 and 0.98 contain two significant figures.
4. Zeros that aren't needed to hold the decimal point are significant. For example, 4.00 has three significant figures.
5. Zeros that follow a number *may* be significant.

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Rules for significant digits:

Rule 1: When approximate numbers are multiplied or divided, the number of significant digits in the final answer is the same as the number of significant digits in the least accurate of the factors.

Example: $X = \frac{45}{(3.22m)x(2.005m)} = 6.97015\ \frac{N}{m^2}$.

Least significant factor (45) has only two (2) digits so only two are justified in the answer. The appropriate way to write the answer is P = 7.0 N/m^2.

Rule 2: When approximate numbers are added or subtracted, the number of significant digits should equal the smallest number of decimal places of any term in the sum or difference.

Example: 9.65 cm + 8.**4** cm - 2.89 cm = 15.**1**6 cm

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1.3. Vectors: composition and resolution

Scalar quantities are measured with numbers and units.

length
(e.g. 16 cm)

temperature
(e.g. 102 °C)

time
(e.g. 7 s)

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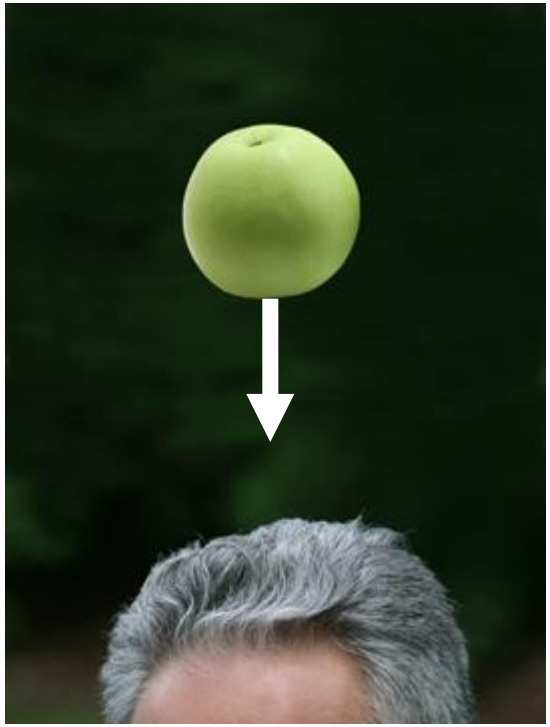
Vector quantities are measured with numbers and units, but also have a specific **direction**.

acceleration

(e.g. 30 m/s^2 upwards)

displacement

(e.g. 200 miles northwest)



force

(e.g. 2 N downwards)

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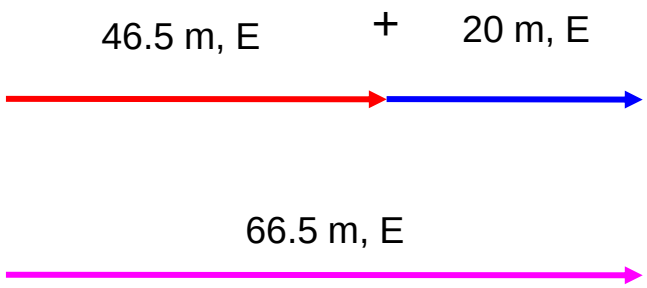
VECTOR APPLICATION

1. ADDITION: When two (2) vectors point in the **SAME** direction, simply add them together.

EXAMPLE: A man walks 46.5 m east, then another 20 m east. Calculate his displacement relative to where he started.



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VECTOR APPLICATION

2. SUBTRACTION: When two (2) vectors point in the **OPPOSITE** direction, simply subtract them.

EXAMPLE: A man walks 46.5 m east, then another 20 m west.

Calculate his displacement relative to where he started.

46.5 m, E

-

20 m, W

26.5 m, E

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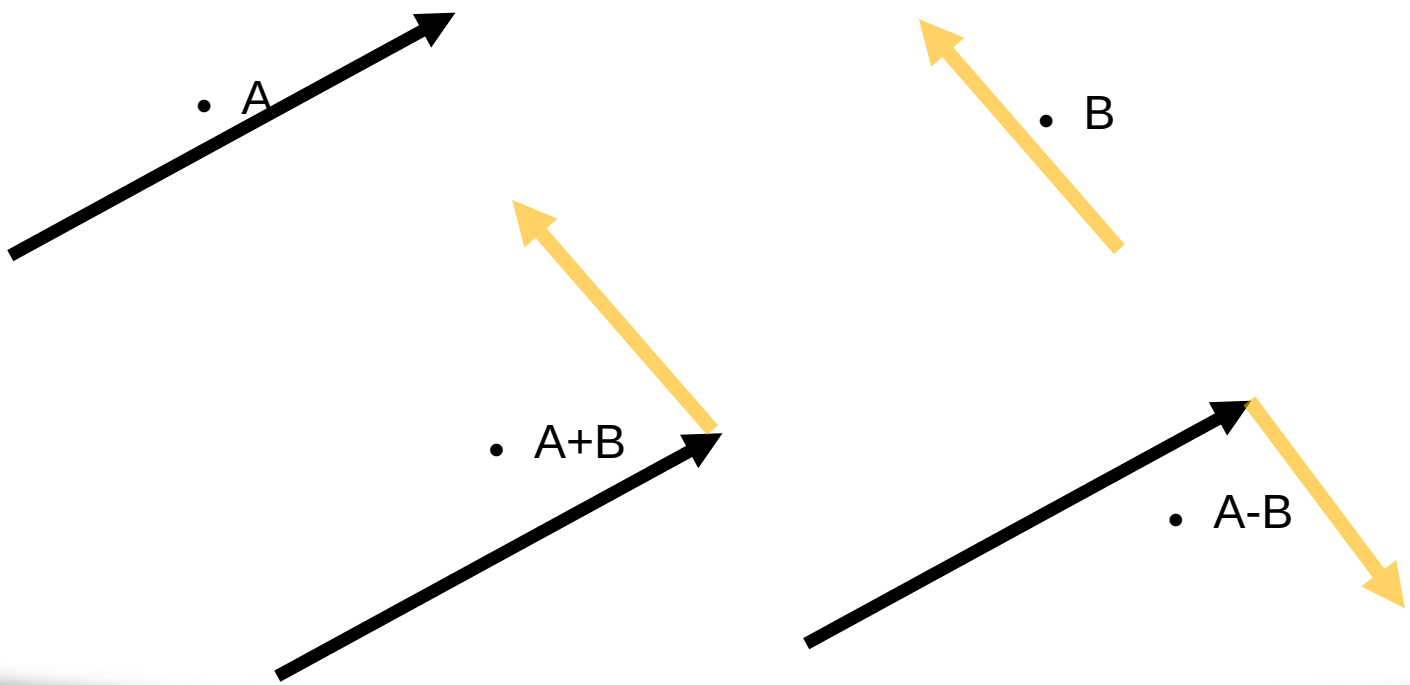
Vectors are always added <u>head</u> to <u>tail</u>

- For example draw what you think it would look like if I added these two vectors together: A+B. Then draw what you think A-B would look like.



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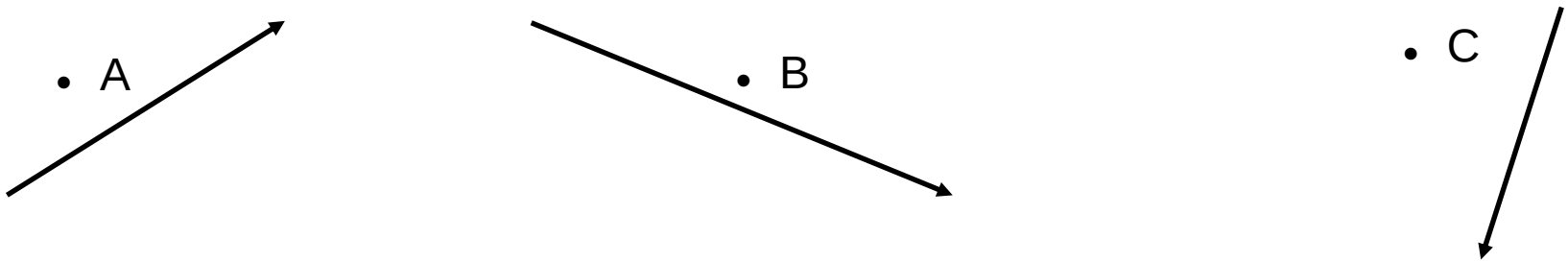
You try on the whiteboard



a. A+C
b. B+C
c. A-B
d. C-B
e. A+B

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NON-COLLINEAR VECTORS

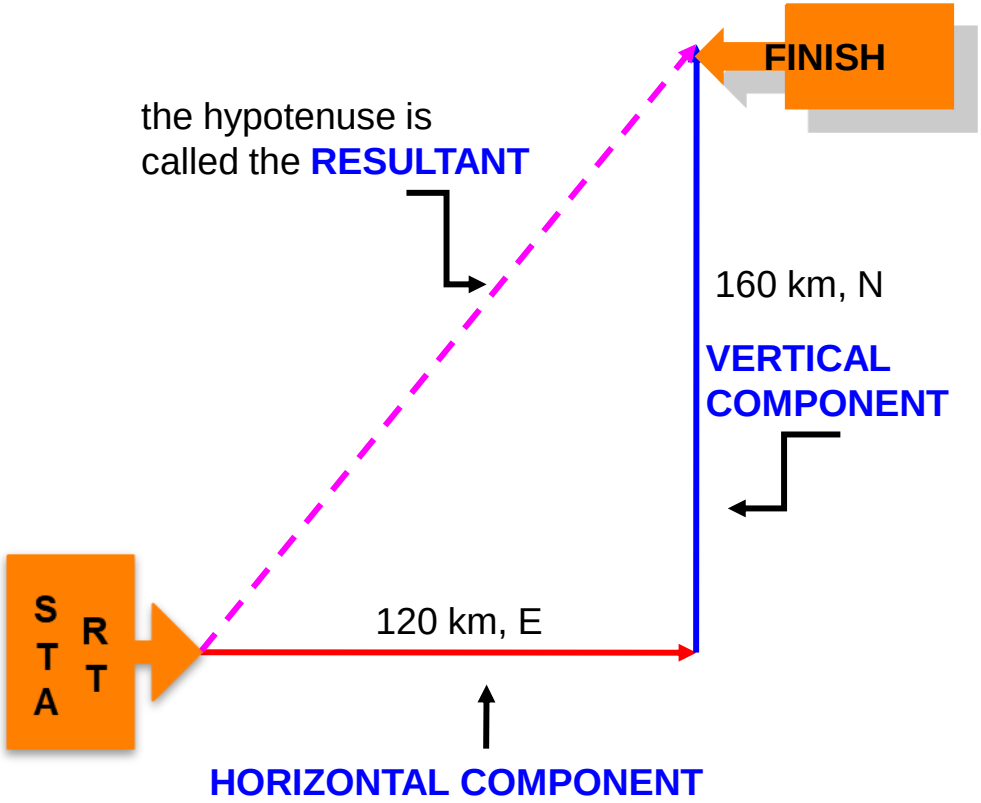
When two (2) vectors are **PERPENDICULAR** to each other, you must use the **PYTHAGOREAN THEOREM**

Example: A man travels 120 km east then 160 km north. Calculate his resultant displacement.

$$c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$$

$$c = \text{resultant} = \sqrt{[(120)^2 + (160)^2]}$$

$$c = 200\text{km}$$



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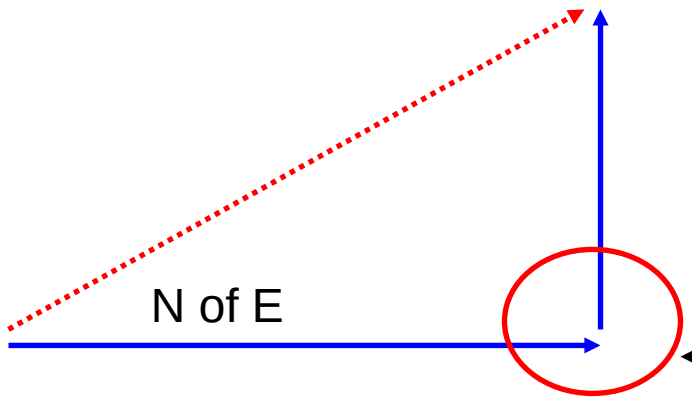
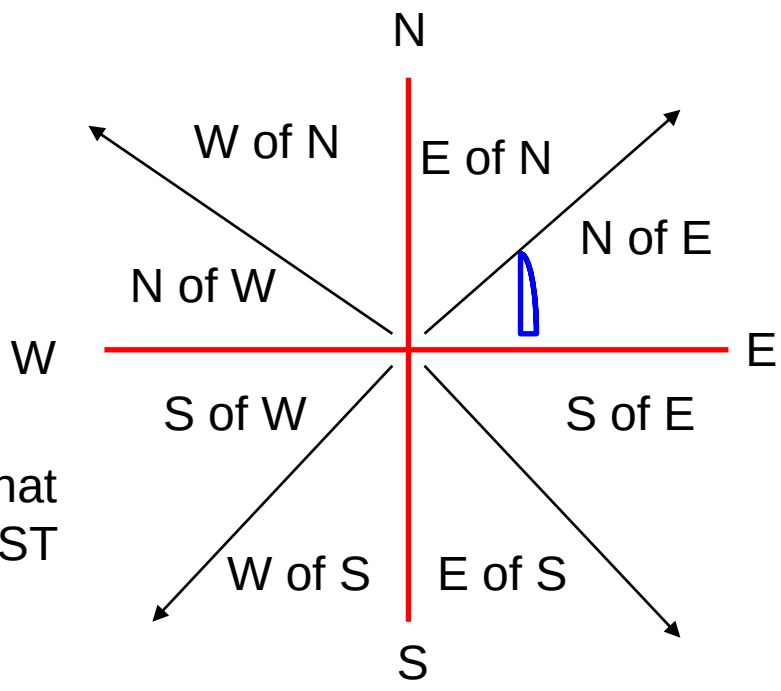
WHAT ABOUT DIRECTION?

In the example, **DISPLACEMENT** asked for and since it is a **VECTOR** quantity, we need to report its direction.





NOTE: When drawing a right triangle that conveys some type of motion, you MUST draw your components **HEAD TO TOE**

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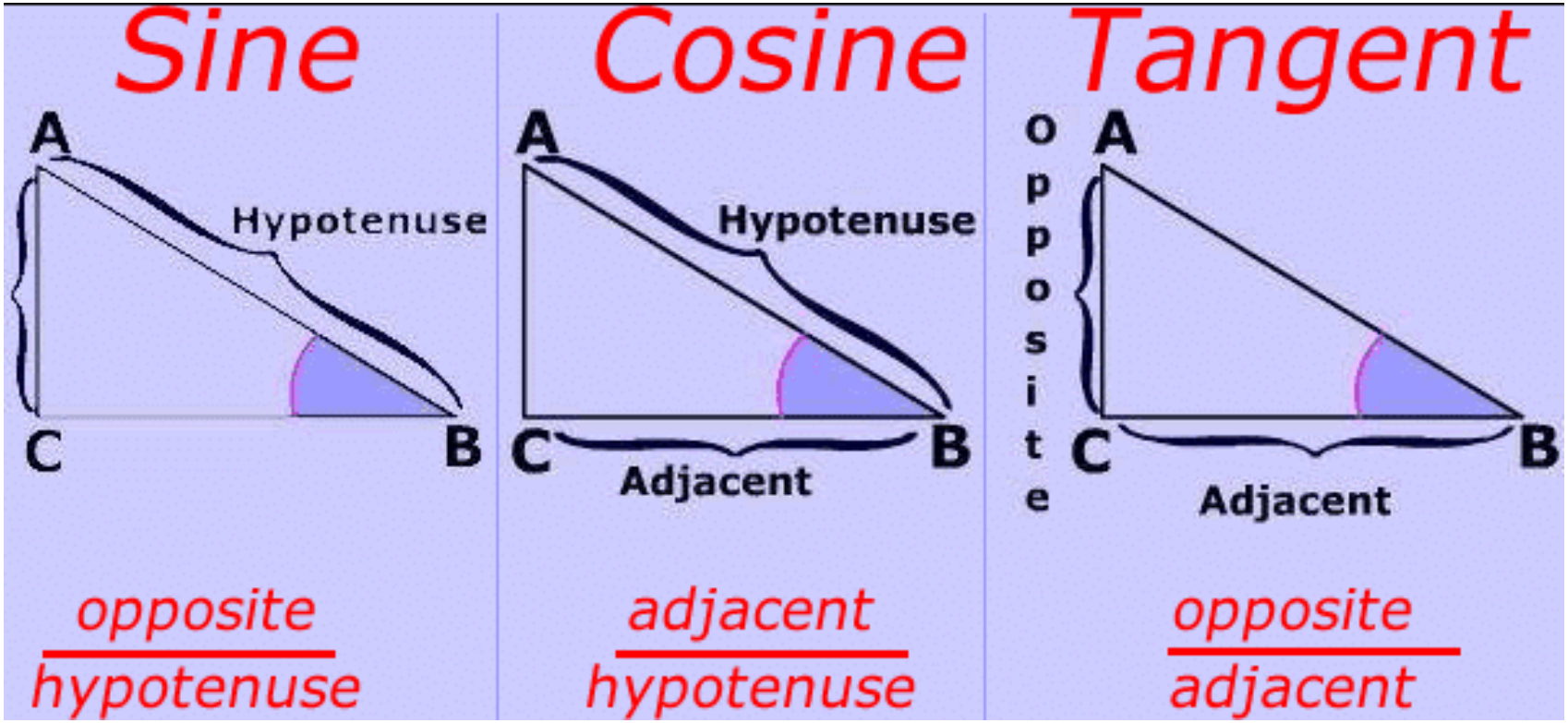
Sine
A
Hypotenuse
C
B
opposite
hypotenuse
Cosine
A
Hypotenuse
C
Adjacent
B
adjacent
hypotenuse
Tangent
Opposite
A
C
Adjacent
B
opposite
adjacent

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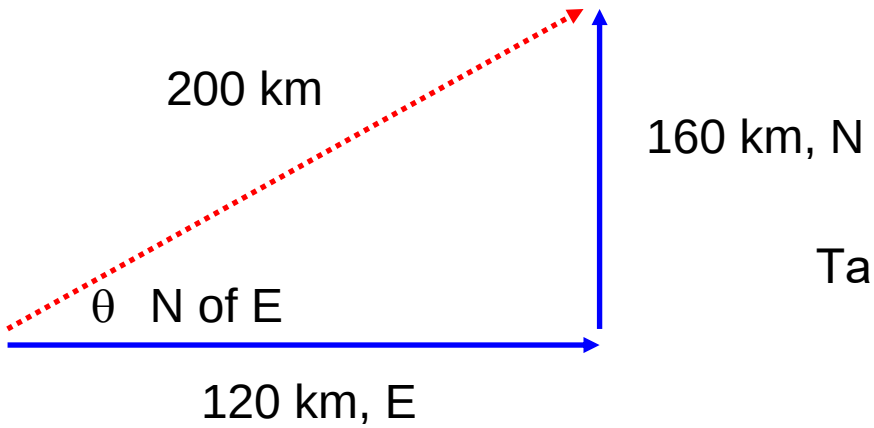
NEED A VALUE – ANGLE !

Just putting N of E is not good enough (how far north of east ?).
We need to find a numeric value for the direction.



To find the value of the angle we use a Trig function called TANGENT.

$$\text{Tan}q = \frac{\text{oppositeside}}{\text{adjacentside}} = \frac{160}{120} = 1.333$$

$$q = \text{Tan}^{-1}(1.333) = 53.1^{o}$$

So the COMPLETE final answer is : **200 km, 53.1 degrees North of East**

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Example-1

1. A bear, searching for food wanders 35 meters east then 20 meters north. Frustrated, he wanders another 12 meters west then 6 meters south. Calculate the bear's displacement.

- Draw the vector diagram
- Combine ‘like’ vectors
- Determine the resultant AND the angle of direction

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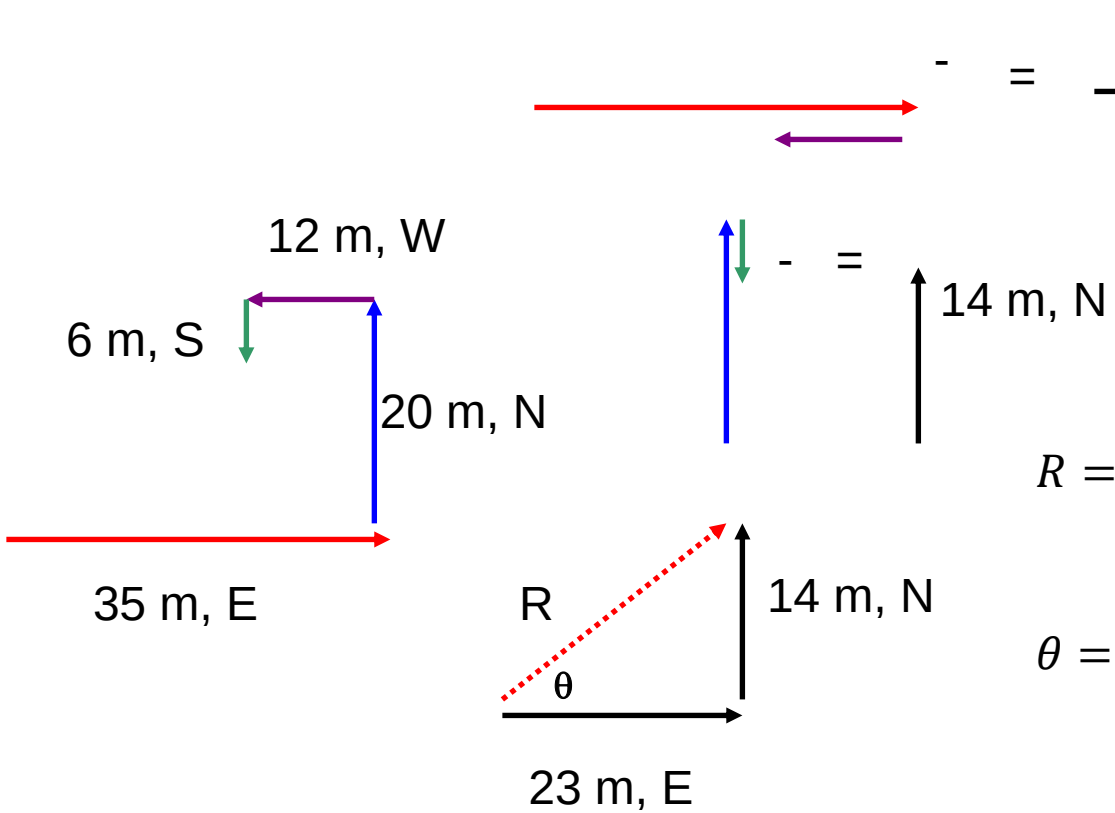
Solution-1

A bear, searching for food wanders 35 meters east then 20 meters north. Frustrated, he wanders another 12 meters west then 6 meters south. Calculate the bear's displacement.



$$R = \sqrt{14^2 + 23^2} = 26.93m$$

$$\text{Tan}\theta = \frac{14}{23} = .6087$$

$$\theta = \text{Tan}^{-1}(0.6087) = 31.3°$$

The Final Answer: **26.93 m, 31.3 degrees NORTH of EAST**

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Example-2

- A boat moves with a velocity of 15 m/s, N in a river which flows with a velocity of 8.0 m/s, west. Calculate the boat's resultant velocity with respect to due north.

- Draw the vector diagram
- Combine ‘like’ vectors (if necessary)
- Determine the resultant AND the angle of direction

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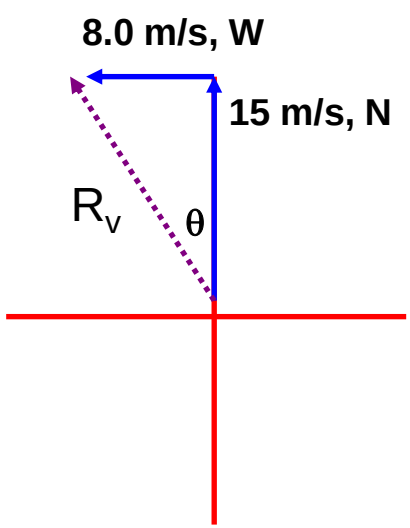
Solution-2

- A boat moves with a velocity of 15 m/s, N in a river which flows with a velocity of 8.0 m/s, west. Calculate the boat's resultant velocity with respect to due north.



$$R_v = \sqrt{8^2 + 15^2} = 17\ m/s$$

$$\text{Tan}\theta = \frac{8}{15} = 0.5333$$

$$\theta = \text{Tan}^{-1}(0.5333) = 28.1^\circ$$

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Example-3

- A plane moves with a velocity of 63.5 m/s at 32 degrees South of East. Calculate the plane's horizontal and vertical velocity components.

- Draw the vector diagram
- Combine ‘like’ vectors (if necessary)
- Determine the resultant AND the angle of direction

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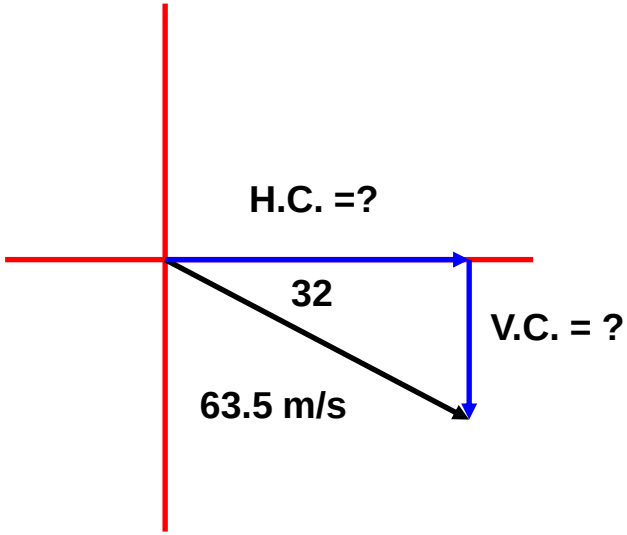
Solution-3

A plane moves with a velocity of 63.5 m/s at 32 degrees South of East.

Calculate the plane's horizontal and vertical velocity components.



$$\text{cosine}\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}, \qquad \text{sine}\theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\text{adj} = \text{hyp.cos}\theta, \qquad \text{opp} = \text{hyp.sin}\theta$$

$$\text{adj} = H.C. = 63.5\cos 32 = 53.85\ m/s\,, E$$

$$\text{opp} = V.C. = 63.5\sin 32 = 33.64\ m/s\,, S$$

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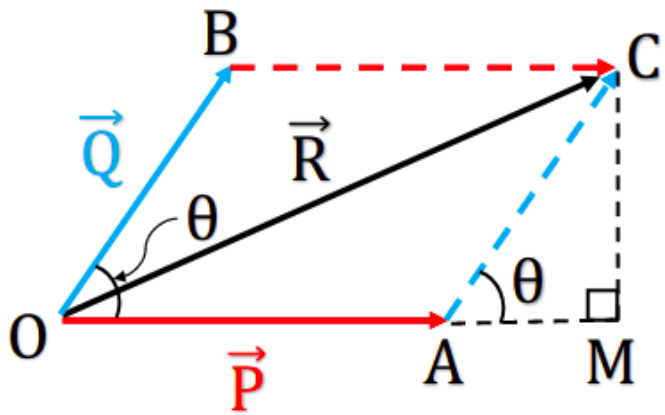
Magnitude of Resultant



In ΔOCM,

$$OC^2 = OM^2 + CM^2$$

$$OC^2 = (OA + AM)^2 + CM^2$$

$$OC^2 = OA^2 + 2OA \times AM + AM^2 + CM^2$$

$$OC^2 = OA^2 + 2OA \times AM + AC^2$$

In ΔCAM,

$$\cos\theta = \frac{AM}{AC} \quad \Rightarrow \quad AM = AC\cos\theta$$

$$OC^2 = OA^2 + 2OA \times AC\cos\theta + AC^2$$

$$R^2 = P^2 + 2P \times Q\cos\theta + Q^2$$

$$\boxed{R = \sqrt{P^2 + 2PQ\cos\theta + Q^2}}$$

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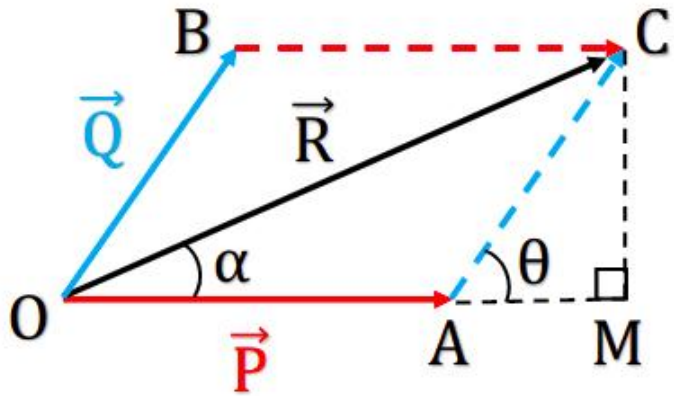
Direction of Resultant



In ΔCAM,

$$\sin\theta = \frac{CM}{AC} \quad \Rightarrow \quad CM = AC\sin\theta$$

$$\cos\theta = \frac{AM}{AC} \quad \Rightarrow \quad AM = AC\cos\theta$$

In ΔOCM,

$$\tan\alpha = \frac{CM}{OM}$$

$$\tan\alpha = \frac{CM}{OA+AM}$$

$$\tan\alpha = \frac{AC\sin\theta}{OA+AC\cos\theta}$$

$$\boxed{\tan\alpha = \frac{Q\sin\theta}{P+Q\cos\theta}}$$

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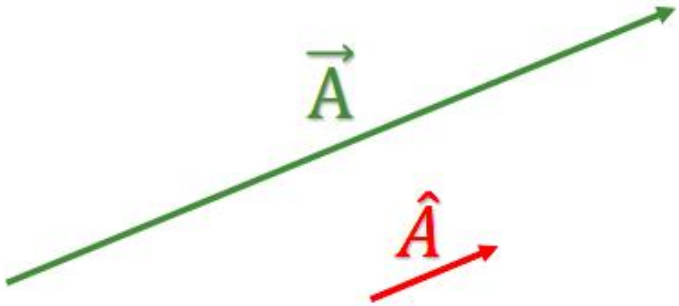
Unit vectors

A unit vector is a vector that has a magnitude of exactly 1 and drawn in the direction of given vector.



- It lacks both dimension and unit.
- Its only purpose is to specify a direction in space.

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Unit vectors

- A given vector can be expressed as a product of its magnitude and a unit vector.
- For example $\vec{A}$ may be represented as,

$$\vec{A} = A\,\hat{A} \quad \Rightarrow \quad \hat{A} = \frac{\vec{A}}{A}$$

A = magnitude of $\vec{A}$

$\hat{A}$ = unit vector along $\vec{A}$

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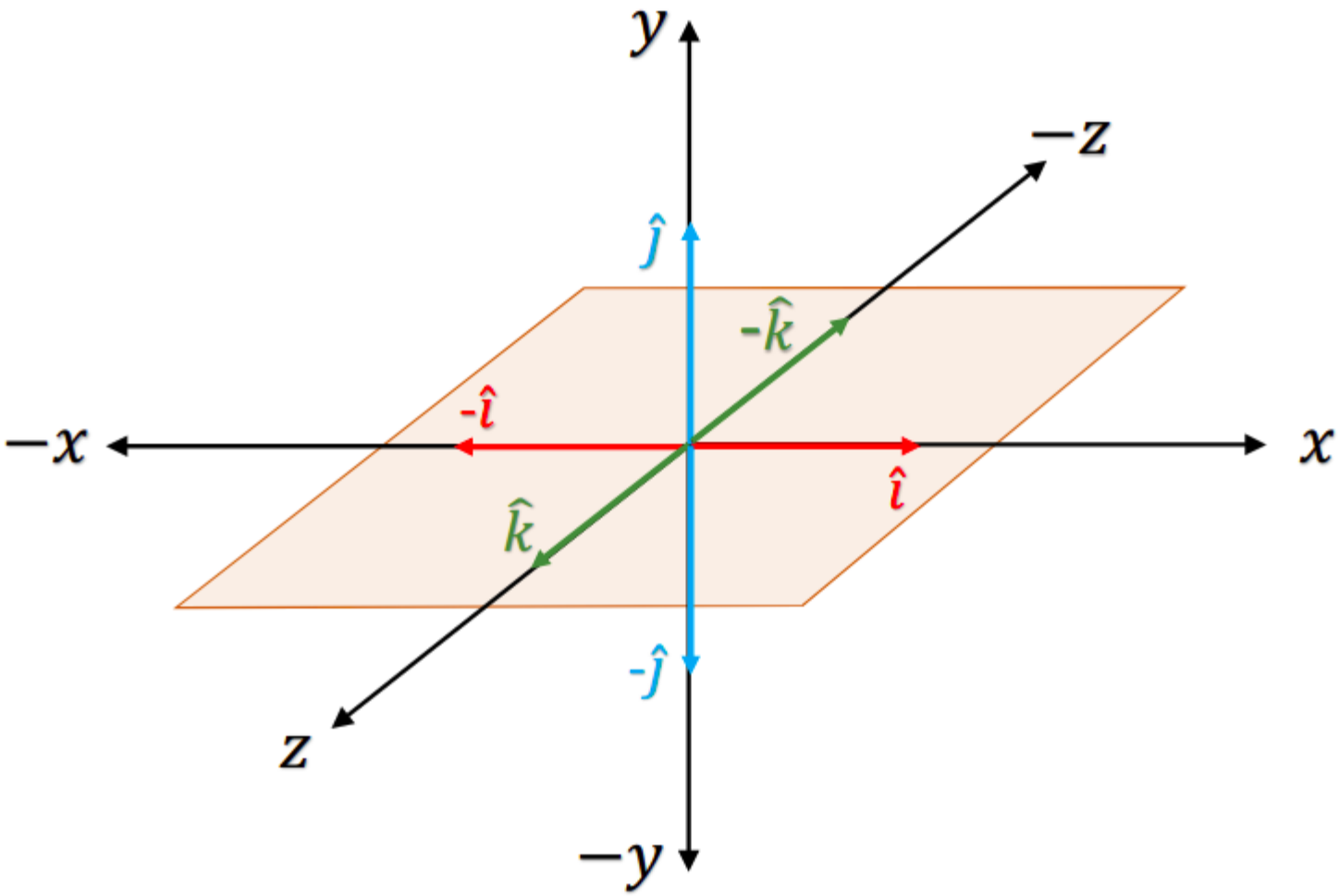
Cartesian unit vectors



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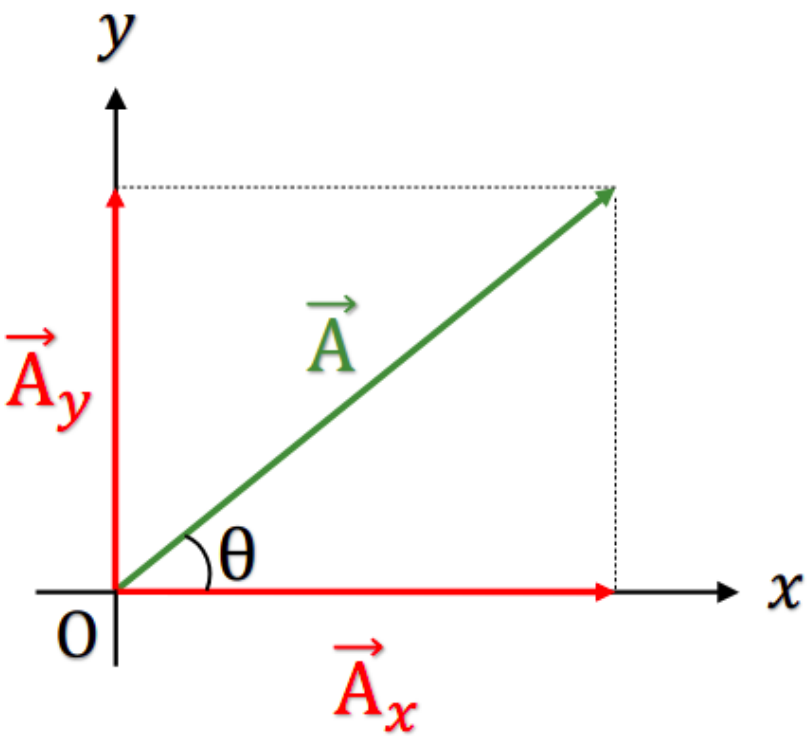
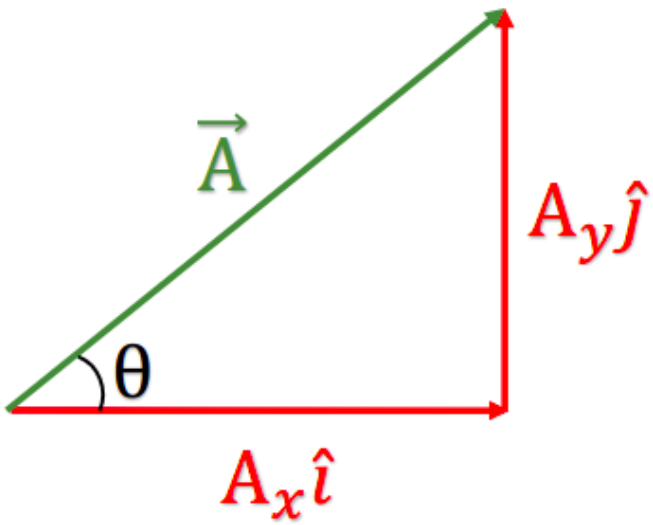
Rectangular Components of 2D Vectors





$$\vec{A} = A_x\hat{i} + A_y\hat{j}$$

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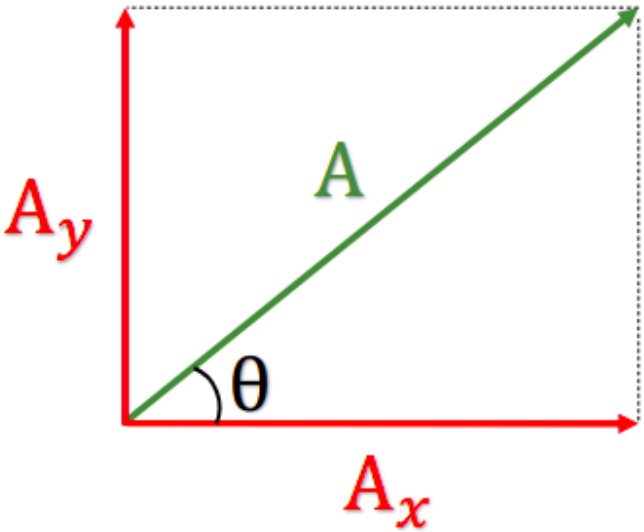
Magnitude & direction from components

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath}$$



Magnitude:

$$A = \sqrt{A_x^2 + A_y^2}$$

Direction:

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

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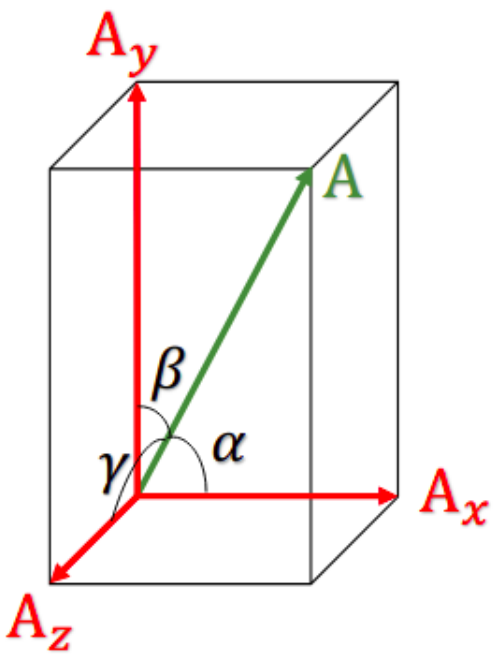
Magnitude & direction from components

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$



Magnitude:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Direction:

$$\alpha = \cos^{-1}\left(\frac{A_x}{A}\right) \qquad \beta = \cos^{-1}\left(\frac{A_y}{A}\right)$$

$$\gamma = \cos^{-1}\left(\frac{A_z}{A}\right)$$

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Adding vectors by components

Let us have

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

then

$$\vec{R} = \vec{A} + \vec{B}$$

$$\vec{R} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k} + B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

$$\vec{R} = (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath} + (A_z + B_z)\hat{k}$$

$$R_x\hat{\imath} + R_y\hat{\jmath} + R_z\hat{k} = (A_x+B_x)\hat{\imath} + (A_y+B_y)\hat{\jmath} + (A_z+B_z)\hat{k}$$

$$R_x = (A_x + B_x)$$

$$R_y = (A_y + B_y)$$

$$R_z = (A_z + B_z)$$

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Multiplying a vector by a scalar

- If we multiply a vector $\vec{A}$ by a scalar s, we get a new vector.
- Its magnitude is the product of the magnitude of $\vec{A}$ and the absolute value of s.
- Its direction is the direction of $\vec{A}$ if s is positive but the opposite direction if s is negative.

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Multiplying a vector by a vector

- There are two ways to multiply a vector by a vector:
- The first way produces a scalar quantity and called as scalar product (dot product).
- The second way produces a vector quantity and called as vector product (cross product).

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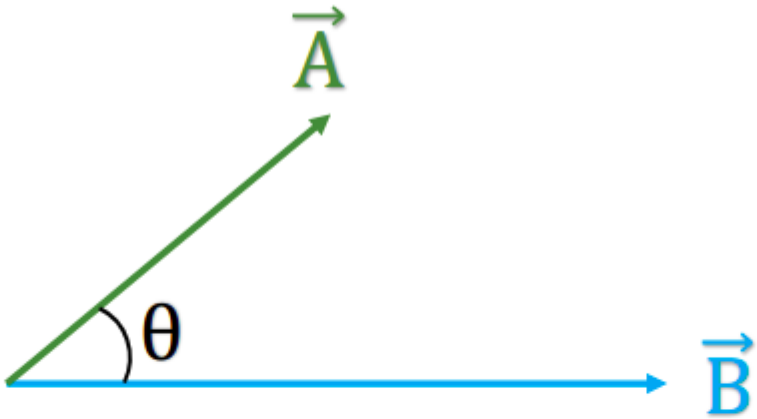
Scalar product



$$\vec{A} \cdot \vec{B} = AB\cos\theta$$

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Examples of scalar product

$$W = \vec{F} \cdot \vec{s}$$

$$W = Fs\cos\theta$$

W = work done

F = force

s = displacement

$$P = \vec{F} \cdot \vec{v}$$

$$P = Fv\cos\theta$$

P = power

F = force

v = velocity

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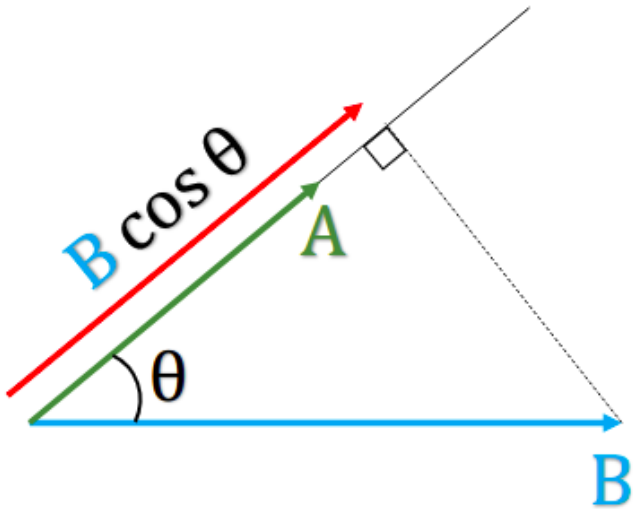
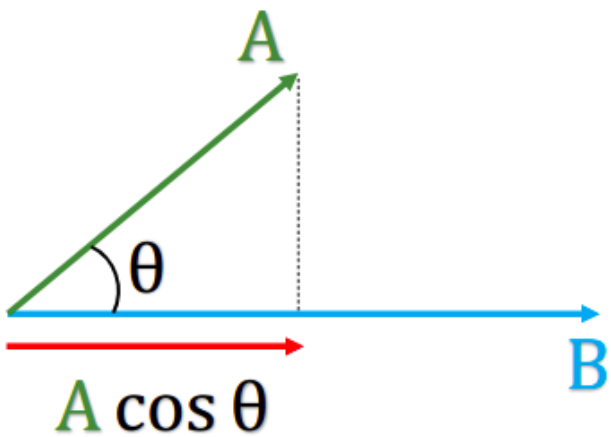
Geometrical meaning of Scalar product

$$\vec{A} \cdot \vec{B} = A(B\cos\theta)$$



$$\vec{A} \cdot \vec{B} = (A\cos\theta)B$$



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Calculating scalar product using components

Let us have

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

then

$$\vec{A}\cdot\vec{B} = \left(A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}\right) \cdot \left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right)$$

$$\begin{aligned}\vec{A}\cdot\vec{B} = {} & A_x\hat{\imath}\cdot\left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right) \\ & + A_y\hat{\jmath}\cdot\left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right) \\ & + A_z\hat{k}\cdot\left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right)\end{aligned}$$

$$\begin{aligned} = {} & A_xB_x\hat{\imath}\cdot\hat{\imath} + A_xB_y\hat{\imath}\cdot\hat{\jmath} + A_xB_z\hat{\imath}\cdot\hat{k} \\ & + A_yB_x\hat{\jmath}\cdot\hat{\imath} + A_yB_y\hat{\jmath}\cdot\hat{\jmath} + A_yB_z\hat{\jmath}\cdot\hat{k} \\ & + A_zB_x\hat{k}\cdot\hat{\imath} + A_zB_y\hat{k}\cdot\hat{\jmath} + A_zB_z\hat{k}\cdot\hat{k}\end{aligned}$$

$$\begin{aligned} = {} & A_xB_x(1) + A_xB_y(0) + A_xB_z(0) \\ & + A_yB_x(0) + A_yB_y(1) + A_yB_z(0) \\ & + A_zB_x(0) + A_zB_y(0) + A_zB_z(1)\end{aligned}$$

$$\boxed{\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z}$$

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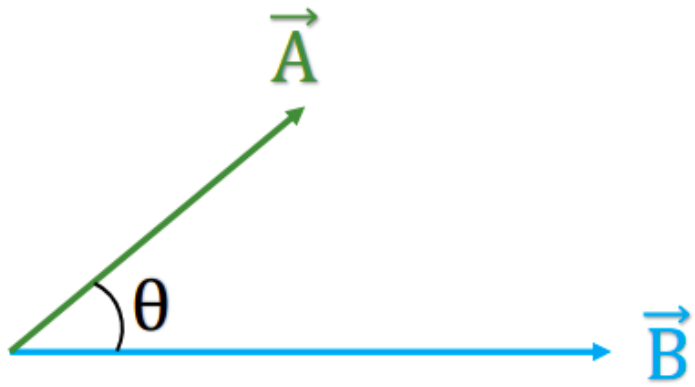
Vector product



$$\vec{A} \times \vec{B} = AB \sin\theta \, \hat{n} = \vec{C}$$

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Examples of vector product

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = rF \sin\theta \, \hat{n}$$

τ = torque

r = position

F = force

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = rp \sin\theta \, \hat{n}$$

L = angular momentum

r = position

p = linear momentum

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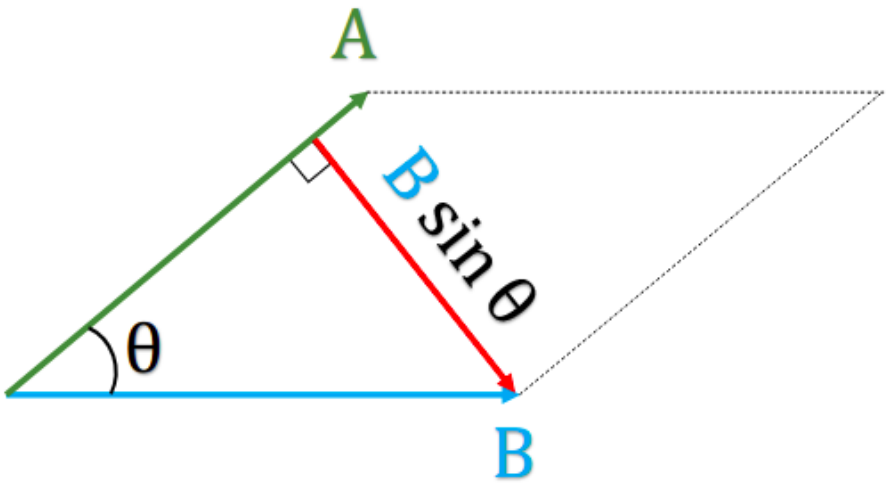
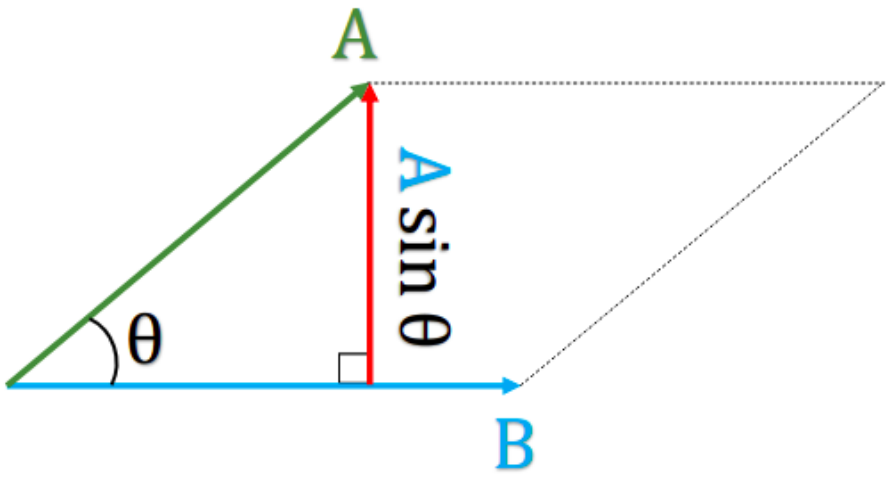
Geometrical meaning of Vector product

$$|\vec{A} \times \vec{B}| = A(B \sin\theta)$$



$$|\vec{A} \times \vec{B}| = (A \sin\theta)B$$



$|\vec{A} \times \vec{B}|$ = Area of parallelogram made by two vectors

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r = position

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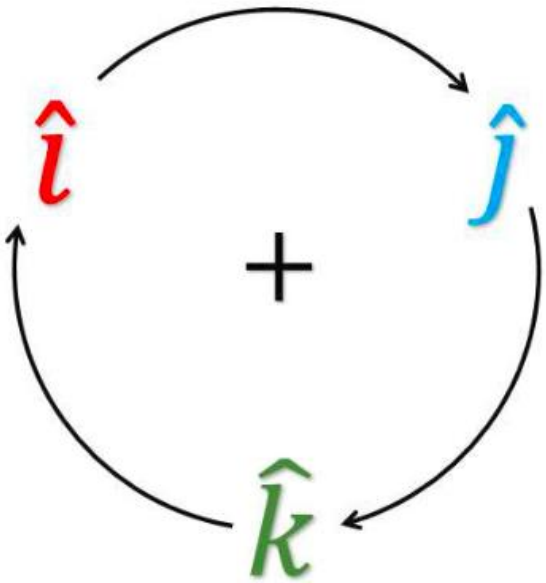
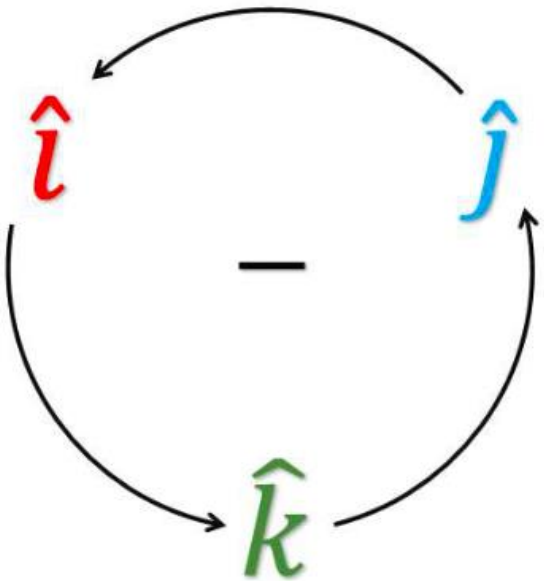
Special Unit Vectors





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Calculating vector product using components

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$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

then

$$\vec{A} \times \vec{B} = \left(A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}\right) \times \left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right)$$

$$\vec{A} \times \vec{B} = A_x\hat{\imath} \times \left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right) + A_y\hat{\jmath} \times \left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right) + A_z\hat{k} \times \left(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}\right)$$

$$= A_xB_x\hat{\imath} \times \hat{\imath} + A_xB_y\hat{\imath} \times \hat{\jmath} + A_xB_z\hat{\imath} \times \hat{k} + A_yB_x\hat{\jmath} \times \hat{\imath} + A_yB_y\hat{\jmath} \times \hat{\jmath} + A_yB_z\hat{\jmath} \times \hat{k} + A_zB_x\hat{k} \times \hat{\imath} + A_zB_y\hat{k} \times \hat{\jmath} + A_zB_z\hat{k} \times \hat{k}$$

$$= A_xB_x(\vec{0}) + A_xB_y(\hat{k}) + A_xB_z(-\hat{\jmath}) + A_yB_x(-\hat{k}) + A_yB_y(\vec{0}) + A_yB_z(\hat{\imath}) + A_zB_x(\hat{\jmath}) + A_zB_y(-\hat{\imath}) + A_zB_z(\vec{0})$$

$$= A_yB_z(\hat{\imath}) - A_zB_y(\hat{\imath}) + A_zB_x(\hat{\jmath}) - A_xB_z(\hat{\jmath}) + A_xB_y(\hat{k}) - A_yB_x(\hat{k})$$

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Calculating vector product using components

$$\vec{A} \times \vec{B} = \hat{\imath}\,(A_y B_z - A_z B_y) - \hat{\jmath}\,(A_x B_z - A_z B_x) + \hat{k}\,(A_x B_y - A_y B_x)$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

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Chapter Review Questions and Problems

1. Vector $\vec{A}$ has magnitude of 8units and makes an angle of 45^0 with the positive x-axis. Vector $\vec{B}$ also has the same magnitude of 8units and directed along the negative x-axis. Find
   a. The magnitude and direction of $\vec{A} + \vec{B}$
   b. The magnitude and direction of $\vec{A} - \vec{B}$
2. Given the displacement vectors $\vec{A} = 3\hat{\imath} - 4\hat{\jmath} + 4\hat{k}$, $\vec{B} = 2\hat{\imath} + 3\hat{\jmath} - 7\hat{k}$. Find the magnitudes of the vectors a) $\vec{A}$+$\vec{B}$ b) 2$\vec{A}$-$\vec{B}$
3. If $\vec{A} = 6\hat{\imath} - 8\hat{\jmath}$, $\vec{B} = -8\hat{\imath} + 3\hat{\jmath}and\vec{C} = 26\hat{\imath} + 19\hat{\jmath}$. Find a and b Such that $a\vec{A}$ +b$\vec{B}$ + $\vec{C} = 0$
4. Find a unit vector in the direction of the resultant of vectors $\vec{A} = 2\hat{\imath} - 3\hat{\jmath} + \hat{k}$, $\vec{B} = \hat{\imath} + \hat{\jmath} + 2\hat{k}$ and $\vec{C} = 3\hat{\imath} - 2\hat{\jmath} + 4\hat{k}$

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$|\vec{A}|$

$$\cos\beta = \frac{A_y}{|\vec{A}|} \Rightarrow \beta = \cos^{-1}\left(\frac{A_y}{|\vec{A}|}\right)$$

$$\underbrace{\cos\gamma = \frac{A_z}{|\vec{A}|}}_{\text{- Directional cosine.}} \Rightarrow \gamma = \cos^{-1}\left(\frac{A_z}{|\vec{A}|}\right)$$

<u>N.B</u>: - Directional cosine $\Rightarrow$ $\cos\alpha = ___$
$\cos\beta = ___$
$\cos\gamma = ___$

- Directional cosine doesn't mean $\alpha = ?$
$\beta = ?$
$\gamma = ?$

<u>Check</u>: Let $\vec{A} = A_x i + A_y j + A_z \hat{k}$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} \Rightarrow |\vec{A}|^2 = A_x^2 + A_y^2 + A_z^2$$

We define:

$$\cos\alpha = \frac{A_x}{|\vec{A}|} \Rightarrow \cos^2\alpha = \frac{A_x^2}{|\vec{A}|^2}$$

$$\cos\beta = \frac{A_y}{|\vec{A}|} \Rightarrow \cos^2\beta = \frac{A_y^2}{|\vec{A}|^2}$$

$$\cos\gamma = \frac{A_z}{|\vec{A}|} \Rightarrow \cos^2\gamma = \frac{A_z^2}{|\vec{A}|^2}$$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma$$

$$= \frac{A_x^2}{|A|^2} + \frac{A_y^2}{|A|^2} + \frac{A_z^2}{|A|^2}$$

$$= \frac{A_x^2 + A_y^2 + A_z^2}{|\vec{A}|^2}$$

$$= \frac{|\vec{A}|^2}{|A|^2}$$

$$= \underline{1}$$

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